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Linear models and Partial Least Square approach

Let (X_i, Y_i) , $i = 1, \dots, n$, be a n -sample, with the $Y_i \in \mathbb{R}$ are outcome variables of interest and the $X_i \in \mathbb{R}^p$ p -dimensional covariates. We consider the linear model

$$Y = X\beta + \varepsilon,$$

where $X = (X_1, \dots, X_n)^T \in \mathbb{R}^{n \times p}$ and $Y = (Y_1, \dots, Y_n)^T \in \mathbb{R}^n$. The noise ε satisfies $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T \sim \mathcal{N}_n(0, \tau^2 I_n)$, with I_n is the identity matrix of size n and $\tau > 0$ the noise standard deviation.

The linear model has been widely investigated, both from practical and theoretical point of view. This project deals with the Partial Least Square (PLS) Algorithm (see for instance [3]). This approach was mainly introduced to deal with the high dimensional case, where $n < p$. The idea is to reduce the number of explanatory variables, by performing a reduction of dimension on the design matrix X . The PLS algorithm seeks for dimensions spanned by the matrix X which are the most correlated with the variable of interest Y .

Logistic linear model

A logistic linear model deals with binary variables of interest $(Y_i)_{i=1, \dots, n}$. Suppose that $Y_i = 0$ or 1 , typically, Y_i is the index of a class belonging. Assume that, for any $i = 1, \dots, n$,

$$Y_i \sim \mathcal{Ber}(\pi_i(\beta)) \quad \text{with} \quad \pi_i(\beta) = \frac{e^{(X\beta)_i}}{1 + e^{(X\beta)_i}}, \quad (1)$$

where $\beta \in \mathbb{R}^p$ denotes the parameter of interest. This parameter is in general estimated via the maximum likelihood principle. Formally, we consider $\hat{\beta}_{ML}$ defined as

$$\hat{\beta}_{ML} = \arg \min_{b \in \mathbb{R}^p} [-l(Y, b)], \quad (2)$$

where $l(Y, b)$ denotes the log-likelihood associated to the sample Y and a parameter $b \in \mathbb{R}^p$. The solution of the optimisation problem (2) is not explicit in this context but optimisation procedures can be deployed. Given an initial value $b^{(0)}$, a current approach is to consider the iterative scheme

$$\begin{cases} z^{(k)} = Xb^{(k)} + (W^{(k-1)})^{-1}(y - \pi^{(k)}), \\ W^{(k)} = \text{diag}(\pi_i(b^{(k)})(1 - \pi_i(b^{(k)})))_{i=1, \dots, p} \\ b^{(k+1)} = (X^T W^{(k-1)} X)^{-1} X^T W^{(k)} z^{(k)}. \end{cases}$$

Provided the algorithm converges, we obtain at the end $\hat{\beta}_{ML} = \beta^\infty$, $\hat{\pi}_{ML} = \pi^\infty$ and $W_\infty = W(\hat{\beta}_{ML})$.

The idea raised in [2] is to use a PLS approach to design an estimator of β from z^∞ .

Objectives

The objectives of the project are to study the quality of the PLS estimation described above on simulations. In particular, we are interested in highlighting the influence on the signal-to-noise ratio on simulations.

An extension can then be to introduce a (Ridge) regularisation to deal with small signal-to-noise ratio cases. First, following [2], a regularization can be included in the iterative resolution of the logistic estimation procedure, before the PLS step. Second, [1] proposes a regularization in the PLS step.

The code can be done with Python or with R.

Bibliography

- [1] L. Castelli, I. Gannaz, and C. Marteau. A non-asymptotic upper bound in prediction for the PLS estimator. Preprint hal-04732627, 2024.
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- [3] A. Höskuldsson. PLS regression methods. *Journal of Chemometrics*, 2(3):211–228, 1988.